

Dynamic Encirclement for Anonymous Agents With Arbitrary Formations

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Abstract—In this study, distributed circle surrounding control problems are investigated for double-integrator anonymous multiagent systems, where all available information for each agent can be obtained based on its moving local coordinate frame. First, the dynamics of each agent described in the local frames are developed in light of the transformation relation between the global and local frames. Then, distributed formation control protocols are designed by local measurements to form and maintain the expected arbitrary circle formations in a plane. The proposed protocols efficiently reduce the requirement of the equipped devices and data processing capabilities, since the agents do not need to be equipped with expensive and high-precision sensors to acquire the real-time acceleration of the target. Finally, the effectiveness of the theoretical result is demonstrated by simulations.

Index Terms—Double-integrator dynamics, formation control, local information, multiagent systems (MASs).

I. INTRODUCTION

A. Background

OVER the last decade, multiagent systems (MASs) have attracted extensive attention because of their broad prospects in civilian and military areas [1], [2], [3], [4]. MASs are composed of multiple agents interacting through a communication network to perform complex tasks. Distributed coordination problems for MASs, where each agent should use only local measurements or interactions to implement the control protocol without a central coordination, have been extensively employed

in traffic control, smart grid, manufacturing, unmanned air vehicle (UAV) formations, sensor networks, data fusion, environmental monitoring, multimanipulator collaborative equipment, and many other fields [5], [6], [7], [8], [9], [10], [11], [12]. Note that formation control is one of the popular applications in the cooperative control for MASs. In such cooperative assignments, the agents are driven to form and maintain the expected formations, as well as to complete complex tasks, such as tracking, exploration, obstacle avoidance, encirclement, and so on [13], [14], [15].

B. Related Papers

Formation control problems have been studied from three perspectives. Following the development of consensus theory, consensus-based formation control problems have been promoted greatly in the past ten years [16], [17], [18], [19], [20], [21]. It should be noted that the acquisition of state information in this context depends on the axis alignment of the coordinate frames of all agents. To overcome the above problem, graph rigidity theory is introduced to accomplish the desired formation by relative distance information without relying on a global or common coordinate system [22], [23], [24], [25]. Different from the above two cases, the general task of the circle formation control problems is usually divided into two subtasks. The first is *target circling*. In other words, each agent keeps the same desired angular velocity around the moving/stationary target and maintains the desired distance from the target. The next is *distribution adjustment*, which means that each agent maintains desired relative angular distances from its two neighbors, respectively. This study focuses on the case of circle formation control. Hence, some existing results on this problem are reviewed in detail in the following context.

In general, circle formation control problems are classified into two cases according to the state variables. In the case of state variables with respect to angles in a circle, most existing results mainly studied the second subobjective. In [26] and [27], sampled-data-based control laws to achieve circle formations were proposed for single- and double-integrator MASs. The follow-up investigations were carried out from different constrained conditions including unidirectional locomotion [28], measurement errors [29], discontinuous updates [30], and limited interaction range [31]. By contrast, both subobjectives can be achieved in the case with taking positions as the state variables in a plane. For single-integrator MASs, Zhao and

Manuscript received 4 June 2023; revised 10 December 2023; accepted 11 December 2023. Date of publication 16 January 2024; date of current version 19 September 2024. This work was supported in part by the Natural Science Foundation of Zhejiang Province under Grant LZ22F030005; in part by the Fundamental Research Funds for the Provincial Universities of Zhejiang under Grant GK239909299001-006; in part by the National Science Foundation of China under Grant 62073108, Grant U22A2044, and Grant 61703130; and in part by the Key Projects of Science and Technology Plan of Zhejiang Province under Grant 2021C01144 and Grant 2019C04018. Recommended by Associate Editor M. Cao. (Corresponding author: Yaguang Kong.)

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Digital Object Identifier 10.1109/TCNS.2024.3354890

Ma [32] constructed a novel controller, which achieves target circling and distribution adjustment in turn. Subsequently, the authors designed new algorithms by employing the key idea of limit cycle to form arbitrarily circle formations for single- and double-integrator MASs [33], [34], [35]. On the above basis, a control protocol with only bearing measurements was designed for single-integrator MASs, but its estimator employs global information [36]. In addition, for single and multiple nonholonomic vehicles, controllers with input saturation to enclose a point were proposed in [37] and [38].

Note that a common and basic assumption in the above studies is that the target's velocity (for single-integrator MASs) or acceleration (for double-integrator MASs) can be accurately measured. In practice, the equipment and data processing capabilities carried by the agents are limited, thus it is almost difficult or impossible to accurately acquire the target's velocity or acceleration information in real time. Therefore, these formation control protocols are difficult to implement in real applications.

C. Contributions

With the above discussions, we mainly focus on the circle surrounding formation problems of double-integrator MASs in this study. Inspired by [35], a novel control scheme is designed to drive each agent to form and maintain desired formations. The contributions of this study are concluded as follows.

- 1) New dynamical equations that describe the agents' dynamics in its local coordinate frame are constructed through transforming from the commonly double-integrator dynamics given in the global coordinate frame. Specifically, all state variables can be acquired by direct measurements or indirect calculations.
- 2) The proposed controllers are totally distributed, which means that only local measurements from each agent's neighbors are required through their local coordinate frames.
- 3) The designed control protocols do not depend on accurate and real-time acceleration measurements of the target. In particular, adaptive laws are constructed such that the parameters of controller design are independent of any global information, such as the upper bound of the target's acceleration.

D. Article Organization

The rest of this article is organized as follows. Notations, definitions on underlying graph, nonsmooth analysis, and problem formulations are presented in Section II. Distributed formation control protocols and the corresponding theoretical analyses are provided in Section III. Numerical simulations are presented in Section IV. Finally, Section V concludes this article.

II. PRELIMINARIES AND PROBLEM FORMULATION

In this section, some notations, graph theory, useful lemmas, and problem formulations are introduced in turn.

A. Common Notations and Graph Theory

Let e denote the exponential constant. $|\cdot|$ is defined as the absolute value. The transpose of matrix M is denoted as M^T . The symbols $\mathbb{R}$, $\mathbb{R}^m$, and $\mathbb{R}^{m \times m}$ represent the 1-D, m -dimensional, and $m \times m$ -dimensional real vector space, respectively. Let notations $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_\infty$ be the 1-, 2-, and ∞ -norm for vectors or matrices, respectively. The symbol $\text{sign}(\cdot)$ denotes the sign function, and let $\text{sign}(\zeta) = [\text{sign}(\zeta_1), \text{sign}(\zeta_2), \dots, \text{sign}(\zeta_n)]^T$, where $\zeta = [\zeta_1, \zeta_2, \dots, \zeta_m]^T \in \mathbb{R}^m$. Let $\wedge$ and $\vee$ denote logical AND and logical OR, respectively. $\mathbf{1}_m$ represents an m -dimensional vector, whose elements are all equal to 1. Let $\text{diag}(\zeta) \in \mathbb{R}^{m \times m}$ be the diagonal matrix constructed from a vector $\zeta \in \mathbb{R}^m$ with the elements in the main diagonal being the elements of vector ζ . For a set S , the symbols $|S|$, $\text{co}(S)$, and $\overline{\text{co}}(S)$ are its cardinality, convex hull, and convex closure, respectively. Define $B(a, r)$ as an open ball with a radius r and a center a . A rectangular coordinate frame can be expressed as $(\bar{e}^x, \bar{e}^y)$, where $\bar{e}^x$ and $\bar{e}^y$ denote its x - and y -axes, respectively. Finally, the notation ∇h is the gradient of a function $h(\cdot)$.

For an MAS with n agents, the communication topology can be described as a undirected graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, v_2, \dots, v_n\}$, and $\mathcal{E}$ are a set of nodes and a set of edges, respectively. An edge $\mathcal{E}_{ij}$ denotes the workable communication between agents i and j . In addition, the communication neighboring set of agent i is denoted as $N_i = \{j \mid (v_i, v_j) \in \mathcal{E}, j \neq i\}$.

B. Definitions and Lemmas

Definition 1 ([39], [40]): A vector field $f[t, \gamma(t)] : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, and define the Filippov set-valued map $\mathcal{K}[f][t, \gamma(t)] \triangleq \bigcap_{\delta > 0} \bigcap_{\mu(N)=0} \overline{\text{co}}(\{f[t, B[\gamma(t), \delta]] - N\})$, where $\bigcap_{\mu(N)=0}$ denotes the intersection over all sets of zero Lebesgue measures.

Definition 2 ([39], [41]): Consider a vector differential equation

$$\dot{\gamma}(t) \in f[t, \gamma(t)] \in \mathbb{R}^n. \quad (1)$$

A vector function $\gamma(\cdot) : \mathbb{R} \rightarrow \mathbb{R}^n$ is called a Filippov solution of (1) on $[t_0, t_1]$, $t_1 \leq \infty$, if $\gamma(\cdot)$ is absolutely continuous and $\dot{\gamma}(t) \in \mathcal{K}[f][t, \gamma(t)]$ for almost all $t \in [t_0, t_1]$.

In addition, conditions on the existence of Filippov solutions are introduced as follows.

Lemma 1 ([39], [41]): Given the system in (1), let $f[t, \gamma(t)]$ be Lebesgue measurable and locally essentially bounded, that is, bounded on a bounded neighborhood of every point, excluding sets of measure zero. Then, for all $\gamma_0 \in \mathbb{R}^n$, there exists a Filippov solution of (1) with the initial condition $\gamma(0) = \gamma_0$.

Definition 3 ([39], [41]): Suppose that $W[\gamma(t)] : \mathbb{R}^n \rightarrow \mathbb{R}$ is a locally Lipschitz function of $\gamma(t)$. The generalized gradient of $W[\gamma(t)]$ is defined as follows:

$$\partial \bar{W}(\gamma) \triangleq \text{co} \left\{ \lim_{w \rightarrow \gamma} \nabla W(w) : w \in \mathbb{R}^n, w \notin \Omega_W \cup S \right\}$$

where Ω_W is the set of points in which $W[\gamma(t)]$ is not differentiable and S is a set of measure zero that can be arbitrarily chosen so as to simplify the calculation. The set-valued Lie derivative

of $W[\gamma(t)]$ with respect to $\gamma(t)$, the trajectory of (1), is defined as $\dot{W}(\gamma) \triangleq \cap_{\xi \in \partial W} \xi^T \mathcal{K}[\mathbf{f}]$.

In the following, a common lemma on the finite-time convergence is reviewed.

Lemma 2 ([42]): Suppose that a nonnegative function $U(t)$ satisfies

$$\dot{U}(t) \leq -\alpha U^p(t), \quad \alpha > 0, \quad 0 < p < 1.$$

Then, $U(t) \equiv 0$ for all $t \geq t_s = \frac{U^{1-p}(t_0)}{\alpha(1-p)}$.

C. Problem Formulation

In an n -agent system, the agents are labeled as 1 to n and the target labeled as 0. This article aims to design locally cooperative control laws for the agents to surround the target in a plane. In the case with an undirected ring graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ among the agents, where $\mathcal{V} = \{1, 2, \dots, n\}$ and $\mathcal{E} = \{(1, 2), (2, 3), \dots, (n-1, n), (n, 1)\}$. Note that the i th agent only has two neighbors described as $N_i = \{i^+, i^-\}$. More specifically, one has the following:

- 1) $i^+ = i + 1$ ($i = 1, 2, \dots, n-1$) and $i^+ = 1$ ($i = n$);
- 2) $i^- = i - 1$ ($i = 2, 3, \dots, n$) and $i^- = n$ ($i = 1$).

We model the dynamics of the i th agent as follows:

$$\begin{cases} \dot{p}_i(t) = v_i(t), \\ \dot{v}_i(t) = u_i(t), \end{cases} \quad i = 1, \dots, n \quad (2)$$

and the dynamics of the target

$$\begin{cases} \dot{p}_0(t) = v_0(t) \\ \dot{v}_0(t) = a_0(t) \end{cases} \quad (3)$$

where $p_i(t) \in \mathbb{R}^2$, $v_i(t) \in \mathbb{R}^2$, and $u_i(t) \in \mathbb{R}^2$ represent the position, velocity, and acceleration of the agents and the target (specially, $u_0(t) \triangleq a_0(t)$) at time t , respectively.

Assumption 1: Assume that $a_0(t) : [0, \infty) \rightarrow \mathbb{R}^2$ is Lebesgue measurable and bounded, i.e., $\|a_0(t)\|_2 \leq \bar{a} \forall t \geq 0$, where $\bar{a}$ is a positive constant and can be unknown.

Denote $\bar{p}_i(t) = p_i(t) - p_0(t)$ and $\bar{v}_i(t) = v_i(t) - v_0(t)$ as the relative position and relative velocity between the agent i and the target, respectively. Thus, one can further obtain the following dynamical system:

$$\begin{cases} \dot{\bar{p}}_i(t) = \bar{v}_i(t) \\ \dot{\bar{v}}_i(t) = -a_0(t) + u_i(t). \end{cases} \quad (4)$$

Let $\alpha_i(t) \in [0, 2\pi)$ be the angular distance from the x -axis in global coordinate frame to vector $\bar{p}_i$ at time t (as shown in Fig. 1). For simplicity to analyze, let

$$0 \leq \alpha_1(0) < \alpha_2(0) < \dots < \alpha_n(0) < 2\pi.$$

Define $\hat{\alpha}_i(t) = \alpha_{i^+}(t) - \alpha_i(t)$ as the angular distance from vector $\bar{p}_{i^+}$ to vector $\bar{p}_i$ at time t and one has

$$\hat{\alpha}_i(0) = \begin{cases} \alpha_{i^+}(0) - \alpha_i(0), & i = 1, 2, \dots, n-1 \\ \alpha_{i^+}(0) - \alpha_i(0) + 2\pi, & i = n. \end{cases}$$

Notice that there always holds $\sum_{i=1}^n \hat{\alpha}_i(0) = 2\pi$. Hence, variables $\hat{\alpha}_i(t)$ are required to be continuous, so that $\sum_{i=1}^n \hat{\alpha}_i(t) = 2\pi$ is always satisfied for arbitrary t .

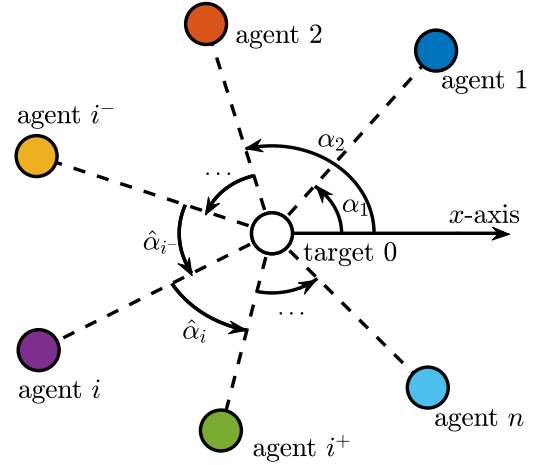


Fig. 1. n agents surround the target 0.

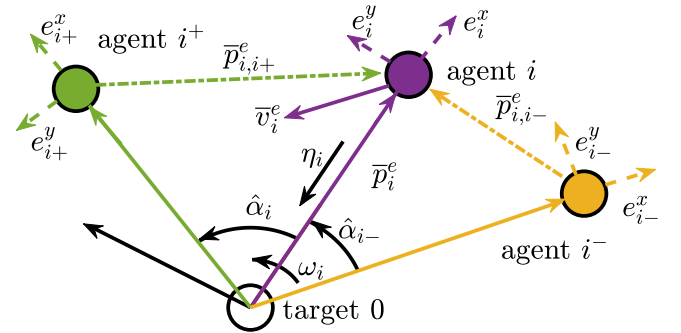


Fig. 2. Measurements and calculations of the state information in (8) for each agent i , where $\bar{p}_{i,i^+}^e$ and $\bar{p}_{i,i^-}^e$ denote the local relative position measurements between agent i and its neighbors i^+ and i^- , respectively.

For each agent i , a moving local coordinate frame is denoted as $(\bar{e}_i^x, \bar{e}_i^y)$ (as shown in Fig. 2), where its x -axis $\bar{e}_i^x$ always points to the position $p_i(t)$ of the i th agent from the position $p_0(t)$ of the target, and its y -axis $\bar{e}_i^y$ is determined via rotating $\pi/2$ (counterclockwise) from the x -axis. Thus, the relationship between the global frame and the moving local frame can be obtained as follows:

$$\begin{cases} u_i^e(t) = A_i^{-1}(t)u_i(t) \\ \bar{v}_i^e(t) = A_i^{-1}(t)\bar{v}_i(t) \\ \bar{p}_i^e(t) = A_i^{-1}(t)\bar{p}_i(t) \\ a_{i0}^e(t) = A_i^{-1}(t)a_0(t) \\ \bar{p}_i^e(t) = [\rho_i(t), 0]^T \end{cases} \quad (5)$$

where $\bar{p}_i^e(t) \in \mathbb{R}^2$, $\bar{v}_i^e(t) \in \mathbb{R}^2$, and $\rho_i(t) = \|\bar{p}_i^e(t)\|_2 > 0$ denote the local relative position, local relative velocity, and distance between agent i and the target, respectively. Let $u_i^e(t) \in \mathbb{R}^2$ be the local input of agent i , and $A_i(t)$ represent the rotation matrix by

$$A_i(t) = \begin{bmatrix} \cos \alpha_i(t) & -\sin \alpha_i(t) \\ \sin \alpha_i(t) & \cos \alpha_i(t) \end{bmatrix}.$$

Based on the last two equations in (5) and the first equation in system (4), one has

$$\begin{aligned}\bar{v}_i &= \dot{\bar{p}}_i \\ &= A_i \dot{\bar{p}}_i^e + \frac{dA_i}{dt} \bar{p}_i^e \\ &= \begin{bmatrix} \cos \alpha_i & -\sin \alpha_i \\ \sin \alpha_i & \cos \alpha_i \end{bmatrix} \begin{bmatrix} \dot{\rho}_i \\ 0 \end{bmatrix} \\ &\quad + \begin{bmatrix} -\dot{\alpha}_i \sin \alpha_i & -\dot{\alpha}_i \cos \alpha_i \\ \dot{\alpha}_i \cos \alpha_i & -\dot{\alpha}_i \sin \alpha_i \end{bmatrix} \begin{bmatrix} \rho_i \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} \dot{\rho}_i \cos \alpha_i - \dot{\alpha}_i \rho_i \sin \alpha_i \\ \dot{\rho}_i \sin \alpha_i + \dot{\alpha}_i \rho_i \cos \alpha_i \end{bmatrix} \\ &= A_i \begin{bmatrix} \dot{\rho}_i \\ \dot{\alpha}_i \rho_i \end{bmatrix}.\end{aligned}$$

Then, one can conclude that

$$\bar{v}_i^e = A_i^{-1}(t) \bar{v}_i(t) = \begin{bmatrix} \eta_i \\ \omega_i \rho_i \end{bmatrix} \quad (6)$$

where $\eta_i = \dot{\rho}_i$ and $\omega_i = \dot{\alpha}_i$.

In addition, from the second equations in (5) and system (4), it follows that:

$$\begin{aligned}u_i &= \dot{\bar{v}}_i + a_0 \\ &= A_i \dot{\bar{v}}_i^e + \frac{dA_i}{dt} \bar{v}_i^e + a_0 \\ &= a_0 + \begin{bmatrix} \cos \alpha_i & -\sin \alpha_i \\ \sin \alpha_i & \cos \alpha_i \end{bmatrix} \begin{bmatrix} \ddot{\rho}_i \\ 0 \end{bmatrix} \\ &\quad + 2 \begin{bmatrix} -\dot{\alpha}_i \sin \alpha_i & -\dot{\alpha}_i \cos \alpha_i \\ \dot{\alpha}_i \cos \alpha_i & -\dot{\alpha}_i \sin \alpha_i \end{bmatrix} \begin{bmatrix} \dot{\rho}_i \\ 0 \end{bmatrix} \\ &\quad + \begin{bmatrix} -\dot{\alpha}_i^2 \cos \alpha_i - \ddot{\alpha}_i \sin \alpha_i & \dot{\alpha}_i^2 \sin \alpha_i - \ddot{\alpha}_i \cos \alpha_i \\ -\dot{\alpha}_i^2 \sin \alpha_i + \ddot{\alpha}_i \cos \alpha_i & -\dot{\alpha}_i^2 \cos \alpha_i - \ddot{\alpha}_i \sin \alpha_i \end{bmatrix} \begin{bmatrix} \rho_i \\ 0 \end{bmatrix} + \\ &= A_i \begin{bmatrix} \ddot{\rho}_i - \dot{\alpha}_i^2 \rho_i \\ 2\dot{\alpha}_i \dot{\rho}_i + \ddot{\alpha}_i \rho_i \end{bmatrix} + a_0.\end{aligned}$$

Thus, it yields

$$u_i^e = A_i^{-1}(t) u_i(t) = \begin{bmatrix} \ddot{\rho}_i - \dot{\alpha}_i^2 \rho_i \\ 2\dot{\alpha}_i \dot{\rho}_i + \ddot{\alpha}_i \rho_i \end{bmatrix} + a_{i0}^e. \quad (7)$$

With the above derivations in (6) and (7), the dynamics of each agent described in the moving local frames are developed as follows:

$$\begin{cases} \dot{\rho}_i = \eta_i \\ \dot{\alpha}_i = \omega_{i+} - \omega_i \\ \dot{\eta}_i = \omega_{i+}^2 \rho_i - a_{i01}^e + u_{i1}^e \\ \dot{\omega}_i = -\frac{1}{\rho_i} (2\omega_i \eta_i + a_{i02}^e) + \frac{1}{\rho_i} u_{i2}^e \end{cases} \quad (8)$$

where $u_i^e = [u_{i1}^e, u_{i2}^e]^T$ and $a_{i0}^e = [a_{i01}^e, a_{i02}^e]^T$.

Assumption 2: Suppose that each agent i can only measure the relative position information of its neighbors and the target as well as the relative velocity information of the target.

Remark 1: By the dynamics in the local frames, we mean that the state information $\rho_i, \eta_i, \omega_i, \dot{\alpha}_i$, and $\dot{\alpha}_{i-}$ for each agent i in (8) can be calculated from local measurements. Note that $\bar{p}_i^e, \bar{v}_i^e, \bar{p}_{i,i+}^e$, and $\bar{p}_{i,i-}^e$ are acquired in agent i 's local coordinate frame (as shown in Fig. 2). Hence, it follows that:

- 1) ρ_i can be obtained directly from the relative position $\bar{p}_i^e = [\rho_i, 0]^T$;
- 2) η_i and ω_i are further determined by the relative velocity $\bar{v}_i^e = [\eta_i, \omega_i \rho_i]^T$ in turn;
- 3) $\dot{\alpha}_i$ and $\dot{\alpha}_{i-}$ are calculated from the relative positions $\bar{p}_{i,i+}^e$ and $\bar{p}_{i,i-}^e$. To be specific, there hold $[\cos \hat{\alpha}_i, \sin \hat{\alpha}_i]^T = \frac{p_i^e - p_{i,i+}^e}{\|p_i^e - p_{i,i+}^e\|_2}$ and $[\cos \hat{\alpha}_{i-}, -\sin \hat{\alpha}_{i-}]^T = \frac{p_i^e - p_{i,i-}^e}{\|p_i^e - p_{i,i-}^e\|_2}$;
- 4) each agent i can obtain its neighbors' angular velocity information ω_{i+} and ω_{i-} via communication interactions.

From Assumption 1, a_{i01}^e and a_{i02}^e are Lebesgue measurable and bounded by

$$\begin{cases} |a_{i01}^e| \leq \bar{a} \\ |a_{i02}^e| \leq \bar{a}. \end{cases} \quad (9)$$

For convenience of follow-up developments, some notations are introduced: $\rho = [\rho_1, \rho_2, \dots, \rho_n]^T$, $\eta = [\eta_1, \eta_2, \dots, \eta_n]^T$, $\hat{\alpha} = [\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_n]^T$, $a_{01}^e = [a_{101}^e, a_{201}^e, \dots, a_{n01}^e]^T$, $a_{02}^e = [a_{102}^e, a_{202}^e, \dots, a_{n02}^e]^T$, as well as

$$\omega_+ = E_+ \omega$$

with $\omega = [\omega_1, \omega_2, \dots, \omega_n]^T$ and

$$E_+ = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \end{bmatrix}.$$

Define the desired angular distance from agent i to its neighbor i^+ as d_i , and the desired distance between agent i and the target as R_i , respectively. Then, let

$$d = [d_1, d_2, \dots, d_n]^T$$

and

$$R = [R_1, R_2, \dots, R_n]^T.$$

In addition, define $\Omega \in \mathbb{R}$ as the desired angular velocity for each agent. Note that each agent rotates counterclockwise (respectively, clockwise) around the target in the desired formation for $\Omega > 0$ (respectively, $\Omega < 0$). In particular, each agent remains stationary relative to the target in the desired formation for $\Omega = 0$. A desired formation (d, R, Ω) is called to be admissible if there hold $\forall R_i > 0 \forall d_i > 0$, and $\sum_{i=1}^n d_i = 2\pi$.

Definition 4 ([35]): Given an admissible formation characterized by $R \in \mathbb{R}^n$ and $d \in \mathbb{R}^n$ in a plane with the desired angular velocity $\Omega \in \mathbb{R}$ to the static/moving target, the desired formation is achieved by designing distributed laws $u_i(t)$ for each agent $i \in \mathcal{V}$, if for any initial states $[p_i^T(t_0), v_i^T(t_0), p_0^T(t_0), v_0^T(t_0)]^T \in$

$\mathbb{R}^8$, $\|a_0(0)\|_2 \leq \bar{a}$, and $\rho_i(t_0) > 0$, the solution of system (8) converges to the desired formation $(\rho^*, \omega^*, \hat{\alpha}^*)$ satisfying

$$\begin{cases} \rho^* = R \\ \omega^* = \Omega \mathbf{1}_n \end{cases} \quad (\text{Target circling}) \quad (10)$$

and

$$\hat{\alpha}^* = d \quad (\text{Distribution adjustment}) \quad (11)$$

where the symbol $\cdot^*$ denotes the desired state.

Lemma 3 ([26]): For a given matrix $L = [l_{ij}]_{n \times n} \in \mathbb{R}^{n \times n}$ with $l_{ii} = \frac{d_{i+}}{d_{i+} + d_i} + \frac{d_{i-}}{d_{i+} + d_{i-}}$, $l_{ii+} = -\frac{d_i}{d_{i+} + d_i}$, $l_{ii-} = -\frac{d_i}{d_{i+} + d_{i-}}$, and $l_{ij} = 0$ for $j \in \mathcal{V} \setminus \{i, i^+, i^-\}$, the eigenvalues of L are denoted as $\lambda_1, \lambda_2, \dots, \lambda_n$, and then the following holds:

- 1) L is diagonalizable and $\lambda_i \in [0, 2]$, $i = 1, 2, \dots, n$;
- 2) 0 is a single eigenvalue;
- 3) when n is even, 2 is an eigenvalue, while when n is odd, 2 is not.

Following Lemma 3, we label that $\lambda_1 = 0 < \lambda_2 \leq \dots \leq \lambda_n \leq 2$.

III. MAIN RESULTS

In this section, local formation control schemes are designed without using the real-time acceleration information of the agents and the target. Then, detailed analyses are developed to prove that the desired formations can be realized under the proposed protocols.

A. Distributed Formation Control Protocols

For the dynamical system (8) in the local frames, a novel formation control strategy for each agent i is proposed as follows:

$$\begin{aligned} u_i^e = \begin{bmatrix} u_{i1}^e \\ u_{i2}^e \end{bmatrix} = \begin{bmatrix} -\omega_i^2 \rho_i - \eta_i \\ (\omega_i + \Omega + f_i) \eta_i + z_i \rho_i \end{bmatrix} \\ + \begin{bmatrix} -l_1 \text{sign}(\rho_i - R_i + \eta_i) \\ l_2 \text{sign}(f_i + \Omega - \omega_i) \end{bmatrix} \end{aligned} \quad (12)$$

where l_1 and l_2 are control parameters to be determined in the following context, and

$$\begin{cases} f_i = \frac{1}{d_{i+} + d_{i-}} (d_{i-} \hat{\alpha}_i - d_i \hat{\alpha}_{i-}) \\ z_i = \frac{1}{d_{i+} + d_{i-}} [d_{i-} (\omega_{i+} - \omega_i) - d_i (\omega_i - \omega_{i-})]. \end{cases} \quad (13)$$

Then, substitute the control law (12) into the system (8) to yield the following compact-form closed-loop system:

$$\begin{cases} \dot{\rho} = \eta \\ \dot{\hat{\alpha}} = \omega_+ - \omega \\ \dot{\eta} = -a_{01}^e \eta - l_1 \text{sign}(\rho - R + \eta) \\ \dot{\omega} = \text{diag}^{-1}(\rho) [-a_{02}^e + \text{diag}(\eta)] (f + \Omega \mathbf{1}_n - \omega) \\ \quad - \text{diag}(\rho) H \omega + l_2 \text{sign}(f + \Omega \mathbf{1}_n - \omega) \end{cases} \quad (14)$$

where $f = [f_1, f_2, \dots, f_n]^T$ and

$$H = \begin{bmatrix} 1 & -\frac{d_n}{d_1 + d_n} & 0 & \dots & 0 & -\frac{d_1}{d_1 + d_n} \\ -\frac{d_2}{d_1 + d_2} & 1 & -\frac{d_1}{d_1 + d_2} & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -\frac{d_{n-1}}{d_n + d_{n-1}} & 0 & 0 & \dots & -\frac{d_n}{d_n + d_{n-1}} & 1 \end{bmatrix}.$$

One can conclude that the existence of the Filippov solutions for system (14) can be guaranteed due to a_{i01}^e , a_{i02}^e , and $\text{sign}(\cdot)$ are Lebesgue measurable and bounded for each agent i .

In addition, we define a sliding-mode variable $s(t) = [s_1^T(t), s_2^T(t)]^T$ for theoretical analyses as follows:

$$\begin{cases} s_1(t) = \rho(t) - R + \eta(t) \\ s_2(t) = \text{diag}(\rho(t))(\omega(t) - \Omega \mathbf{1}_n - f(t)). \end{cases} \quad (15)$$

Then, a theorem can be established for achieving desired formations by analyzing the closed-loop system (14) via the Lyapunov stability theory as follows.

Theorem 1: Suppose that Assumptions 1 and 2 are satisfied, and the upper bound $\bar{a}$ is known. Then, for the multiagent system (8) with protocol (12), the following holds:

- 1) the sliding-mode surface $s = 0$ can be reached within a finite time $t_s \triangleq \frac{\|s(0)\|_2}{l - n\bar{a}}$, where $l = \min\{l_1, l_2\}$ with $l_i > n\bar{a}$, $i = 1, 2$;
- 2) on the sliding-mode surface, the solution of system (8) converges to the desired formation $(\rho^*, \omega^*, \hat{\alpha}^*)$ at an exponential rate $\min\{1, \lambda_2\}$.

Proof: The proof is divided into two parts. First, we construct a Lyapunov function candidate $V(t)$ and prove the sliding-mode surface tends to zero in a finite settling time. Then, we present the final formation performances.

Step 1: Consider the Lyapunov function candidate as follows:

$$V(t) = \frac{1}{2} s^T(t) s(t).$$

For any variable $q(t) \in \mathbb{R}$, the Filippov set-valued map is defined as

$$\mathcal{K}[\text{sign}(q(t))] = \begin{cases} \{1\}, & q(t) > 0 \\ [-1, 1], & q(t) = 0 \\ \{-1\}, & q(t) < 0. \end{cases}$$

In what follows, two cases are considered for analyses.

Case i: When $(\rho_i(t) - R_i + \eta_i(t) \neq 0) \wedge (\omega_i(t) - \Omega - f_i(t) \neq 0)$ for $\forall i \in \mathcal{V}$. In this case, one drives that

$$\begin{cases} \mathcal{K}[\text{sign}(\rho_i(t) - R_i + \eta_i(t))] = \{\text{sign}(\rho_i(t) - R_i + \eta_i(t))\} \\ \mathcal{K}[\text{sign}(\omega_i(t) - \Omega - f_i(t))] = \{\text{sign}(\omega_i(t) - \Omega - f_i(t))\}. \end{cases}$$

The generalized gradients of $V(t)$ with respect to $s_1(t)$ and $s_2(t)$ are calculated as follows:

$$\frac{\partial \tilde{V}(t)}{\partial s_1(t)} = s_1(t), \quad \frac{\partial \tilde{V}(t)}{\partial s_2(t)} = s_2(t)$$

and then the set-valued Lie derivative of $V(t)$ along with (14) is given by

$$\begin{aligned}
\dot{V} &\in \left(\frac{\partial \tilde{V}}{\partial s_1} \right)^T \mathcal{K}[\dot{s}_1] + \left(\frac{\partial \tilde{V}}{\partial s_2} \right)^T \mathcal{K}[\dot{s}_2] \\
&\subseteq (\rho - R + \eta)^T (\eta + \mathcal{K}[\dot{\eta}]) \\
&\quad + (\omega - \Omega \mathbf{1}_n - f)^T \text{diag}(\rho) \text{diag}(\rho) (\mathcal{K}[\dot{\omega}] + H\omega) \\
&\quad + (\omega - \Omega \mathbf{1}_n - f)^T \text{diag}(\rho) \text{diag}(\eta) (\omega - \Omega \mathbf{1}_n - f) \\
&= (\rho - R + \eta)^T (\eta - \eta - a_{01}^e - l_1 \{\text{sign}(\rho - R + \eta)\}) \\
&\quad + (\omega - \Omega \mathbf{1}_n - f)^T \text{diag}(\rho) [\text{diag}(\eta) (f + \Omega \mathbf{1}_n - \omega) \\
&\quad + \text{diag}(\eta) (\omega - \Omega \mathbf{1}_n - f) - \text{diag}(\rho) H\omega \\
&\quad + \text{diag}(\rho) H\omega - a_{02}^e + l_2 \{\text{sign}(f + \Omega \mathbf{1}_n - \omega)\}] \\
&= (\rho - R + \eta)^T (-a_{01}^e - l_1 \text{sign}(\rho - R + \eta)) \\
&\quad + (\omega - \Omega \mathbf{1}_n - f)^T \text{diag}(\rho) \\
&\quad \times \{-a_{02}^e + l_2 \text{sign}[\text{diag}(\rho) (f + \Omega \mathbf{1}_n - \omega)]\}
\end{aligned}$$

where the fact $\{\text{sign}(f + \Omega \mathbf{1}_n - \omega)\} = \{\text{sign}[\text{diag}(\rho) (f + \Omega \mathbf{1}_n - \omega)]\}$ is employed as $\rho_i > 0$ for each agent i . Then, it follows from inequalities (9) and $l_i > n\bar{a}$ ($i = 1, 2$) that

$$\begin{aligned}
\dot{V} &\leq -(l_1 - n\bar{a}) \|\rho - R + \eta\|_1 \\
&\quad - (l_2 - n\bar{a}) \|\text{diag}(\rho) (f + \Omega \mathbf{1}_n - \omega)\|_1 \leq 0. \quad (16)
\end{aligned}$$

Case ii: There exists an agent i such that $(\rho_i(t) - R_i + \eta_i(t) = 0) \vee (\omega_i(t) - \Omega - f_i(t) = 0)$. In a similar trace, one can obtain

$$\begin{aligned}
\dot{V} &\leq -(l_1 - n\bar{a}) \|\rho - R + \eta\|_1 \\
&\quad - (l_2 - n\bar{a}) \|\text{diag}(\rho) (f + \Omega \mathbf{1}_n - \omega)\|_1 \leq 0.
\end{aligned}$$

As a consequence, from Cases i) and ii), one can conclude that

$$\begin{aligned}
\dot{V} &\leq -(l_1 - n\bar{a}) \|s_1\|_1 - (l_2 - n\bar{a}) \|s_2\|_1 \\
&\leq -(l - n\bar{a}) (\|s_1\|_2 + \|s_2\|_2) \\
&\leq -(l - n\bar{a}) \sqrt{\|s_1\|_2^2 + \|s_2\|_2^2} \\
&= -\sqrt{2}(l - n\bar{a}) V^{\frac{1}{2}}. \quad (17)
\end{aligned}$$

In light of Lemma 2, the inequality (17) indicates that $V(t)$ tends to zero in a finite settling time t_s , which means that $s(t) \rightarrow 0$ as $t \rightarrow t_s$.

Step 2: In the following context, we discuss the formation performances under the proposed protocol (12).

Given that $s_1(t) = 0$ and $s_2(t) = 0$ for $\forall t \geq t_s$ from Step 1, following from (15), it yields

$$\begin{cases} \eta(t) = R - \rho(t) \\ \omega(t) - \Omega \mathbf{1}_n = f(t). \end{cases} \quad (18)$$

Based on the first two equations in system (14), for $t \geq t_s$, one can drive in light of [26, Th. 1] that

$$\begin{cases} \dot{\rho}(t) = R - \rho(t) \\ \dot{\hat{\alpha}}(t) = -L\hat{\alpha}(t) \end{cases} \quad (19)$$

where the matrix L is given in Lemma 3.

Thus, one can get

$$\begin{cases} \rho(t) = e^{-(t-t_s)} (\rho(t_s) - R) + R \\ \hat{\alpha}(t) = e^{-L(t-t_s)} \hat{\alpha}(t_s) \end{cases} \quad (20)$$

which implies that

$$\lim_{t \rightarrow \infty} \rho(t) \rightarrow R$$

and

$$\lim_{t \rightarrow \infty} \hat{\alpha}(t) \rightarrow d.$$

Recalling from (13), it is easy to obtain that

$$\lim_{t \rightarrow \infty} f(t) \rightarrow 0.$$

Then, following the second equation in (18), one has

$$\lim_{t \rightarrow \infty} \omega \rightarrow \Omega \mathbf{1}_n.$$

In addition, from (20) and Lemma 3, the agents can realize the desired formation at an exponential rate determined by $\min\{1, \lambda_2\}$. The above completes the proof. $\square$

Remark 2: Compared with the existing protocols designed in [35], which require the real-time and accurate acceleration information $a_0^e(t)$ of the target measured in each agent's local coordinate system, only the controller parameters l_1 and l_2 in (12) are determined by the upper bound of $a_0^e(t)$ instead of its actual values. This indicates that the proposed protocol (12) is robust to the acceleration information $a_0^e(t)$. In addition to this, adaptive laws for the controller parameters are developed that do not involve any information related to the acceleration of the target in the following section.

Remark 3: Note that the existing protocol proposed in [43] utilizes the angle information $\alpha_i(t)$ of agent i obtained in the global coordinate system (as shown in Fig. 1). By contrast, protocol (12) is not dependent on $\alpha_i(t)$; instead, the relative angle information $\hat{\alpha}_i(t)$ calculated from local information (recalling it from Remark 1) is utilized for the control design.

In the above developments, the designed control protocols rely on the upper bound $\bar{a}$ of the target's acceleration $a_0^e(t)$. However, in some practical situations, $\bar{a}$ is often difficult to be obtained for each agent i in advance. Furthermore, once the corresponding condition changes, the coupling gains l_1 and l_2 are required to be reset. Accordingly, new strategies are proposed for the controller design to overcome these limitations in the following.

B. Distributed Adaptive Formation Control Protocols

We construct the following distributed controllers with adaptive laws for updating the control parameters to solve the above

concerns:

$$u_i^e = \begin{bmatrix} u_{i1}^e \\ u_{i2}^e \end{bmatrix} = \begin{bmatrix} -\omega_i^2 \rho_i - \eta_i \\ (\omega_i + \Omega + f_i)\eta_i + z_i \rho_i \end{bmatrix} + \begin{bmatrix} -e_{i1} \text{sign}(\rho_i - R_i + \eta_i) \\ e_{i2} \text{sign}(f_i + \Omega - \omega_i) \end{bmatrix} \quad (21)$$

and

$$\begin{cases} \dot{e}_{i1} = \tau_{i1} \|\rho_i - R_i + \eta_i\|_1 \\ \dot{e}_{i2} = \tau_{i2} \|\rho_i(\omega_i - \Omega - f_i)\|_1 \end{cases} \quad (22)$$

where f_i and z_i are given in (13), and $\tau_{i1} > 0$ and $\tau_{i2} > 0$ are the constants.

Denote $e_1 = [e_{11}, e_{21}, \dots, e_{n1}]^T$, and $e_2 = [e_{12}, e_{22}, \dots, e_{n2}]^T$. Then, for system (8) with adaptive control protocol (21), the compact-form closed-loop system can be described as follows:

$$\begin{cases} \dot{\rho} = \eta \\ \dot{\hat{\alpha}} = \omega_+ - \omega \\ \dot{\eta} = -a_{01}^e - \eta - \text{diag}(e_1) \text{sign}(\rho - R + \eta) \\ \dot{\omega} = \text{diag}^{-1}(\rho)[-a_{02}^e + \text{diag}(\eta)(f + \Omega \mathbf{1}_n - \omega) \\ \quad - \text{diag}(\rho)H\omega + \text{diag}(e_2) \text{sign}(f + \Omega \mathbf{1}_n - \omega)]. \end{cases} \quad (23)$$

In this context, we can obtain the following results.

Theorem 2: Suppose that Assumptions 1 and 2 are satisfied, then the multiagent system (8) with protocol (21) asymptotically achieves the desired formation $(\rho^*, \omega^*, \hat{\alpha}^*)$.

Proof: Take the following Lyapunov function candidate:

$$V_2 = \frac{1}{2} s_1^T s_1 + \frac{1}{2} s_2^T s_2 + \sum_{i=1}^n \frac{1}{2\tau_{i1}} (e_{i1} - r_1)^2 + \sum_{i=1}^n \frac{1}{2\tau_{i2}} (e_{i2} - r_2)^2 \quad (24)$$

where r_1 and r_2 are the positive constants.

Similar to the proof of Theorem 1, one has

$$\begin{aligned} \dot{V}_2 &\in (\rho - R + \eta)^T \{-a_{01}^e - \text{diag}(e_1) \text{sign}(\rho - R + \eta)\} \\ &\quad + (\omega - \Omega \mathbf{1} - f)^T \text{diag}(\rho) \\ &\quad \times \{-a_{02}^e + \text{diag}(e_2) \text{sign}[\text{diag}(\rho)(f + \Omega \mathbf{1}_n - \omega)]\} \\ &\quad + \sum_{i=1}^n \frac{\dot{e}_{i1}(e_{i1} - r_1)}{\tau_{i1}} + \sum_{i=1}^n \frac{\dot{e}_{i2}(e_{i2} - r_2)}{\tau_{i2}} \\ &\leq n\bar{a} \|\rho - R + \eta\|_1 \\ &\quad + n\bar{a} \|\text{diag}(\rho)(\omega - \Omega \mathbf{1} - f)\|_1 \\ &\quad - \sum_{i=1}^n e_{i1} \left(\|\rho_i - R_i + \eta_i\|_1 - \frac{\dot{e}_{i1}}{\tau_{i1}} \right) \\ &\quad - \sum_{i=1}^n e_{i2} \left(\|\text{diag}(\rho)(\omega - \Omega \mathbf{1} - f)\|_1 - \frac{\dot{e}_{i2}}{\tau_{i2}} \right) \\ &\quad - \sum_{i=1}^n \frac{\dot{e}_{i1}}{\tau_{i1}} r_1 - \sum_{i=1}^n \frac{\dot{e}_{i2}}{\tau_{i2}} r_2 \end{aligned}$$

$$\begin{aligned} &= -(r_1 - n\bar{a}) \|\rho - R + \eta\|_1 \\ &\quad - (r_2 - n\bar{a}) \|\text{diag}(\rho)(\omega - \Omega \mathbf{1} - f)\|_1 \\ &= -(r_1 - n\bar{a}) \|s_1\|_1 - (r_2 - n\bar{a}) \|s_2\|_1. \end{aligned}$$

By selecting sufficiently large r_1 and r_2 such that $r_1 > n\bar{a}$ and $r_2 > n\bar{a}$ are satisfied, it yields

$$\begin{aligned} \dot{V}_2 &\leq -(r_1 - n\bar{a}) \|s_1\|_1 - (r_2 - n\bar{a}) \|s_2\|_2 \\ &\triangleq -W(s_1, s_2) \leq 0 \end{aligned}$$

which implies that $V_2(t)$ is monotonically decreasing and bounded with $V_2(t) \rightarrow V_2^\infty$ as $t \rightarrow \infty$. Thus, it follows from the Lyapunov function (24) that e_{i1} , e_{i2} , s_1 , and s_2 are bounded. Also, one can obtain that $\int_0^\infty W(s_1(\tau), s_2(\tau)) d\tau$ exists and is bounded since $\int_0^\infty W(s_1(\tau), s_2(\tau)) d\tau \leq V_2(s_1(t_0), s_2(t_0)) - V_2^\infty$. In addition, $\dot{s}_1 = -a_{01}^e - \text{diag}(e_1) \text{sign}(\rho - R + \eta)$ and $\dot{s}_2 = -a_{02}^e + \text{diag}(e_2) \text{sign}[\text{diag}(\rho)(f + \Omega \mathbf{1}_n - \omega)]$ are bounded from Assumption 1, which in turn implies that $\dot{W}(s_1, s_2)$ is bounded. Accordingly, in light of Barbalat's Lemma (see [44, Lemma A.4.7]), we can conclude that $W(s_1(t), s_2(t)) \rightarrow 0$ as $t \rightarrow \infty$, i.e., $\lim_{t \rightarrow \infty} s_1(t) = 0$ and $\lim_{t \rightarrow \infty} s_2(t) = 0$. Then, from (22), e_{i1} and e_{i2} converge to some finite values, due to the boundedness of e_{i1} and e_{i2} , as well as $\tau_{i1} > 0$ and $\tau_{i2} > 0$.

In retrospect, from the sliding-mode variable defined in (15) and the first two equations in (23) one can obtain

$$\begin{cases} \dot{\rho}(t) = R - \rho(t) + s_1(t) \\ \dot{\hat{\alpha}}(t) = -L\hat{\alpha}(t) + g(t) \end{cases} \quad (25)$$

where $g(t) = y_+(t) - y(t)$ with $y(t) = \text{diag}^{-1}(\rho(t))s_2(t)$ and $y_+(t) = E_+ y(t)$ [where E_+ is given in (14)].

Note that the system (25) is input-to-state stable ([45, Def. 4.7]) with respect to the inputs $s_1(t)$ and $g(t)$ as well as the states $\rho(t)$ and $\hat{\alpha}(t)$. Following from $\lim_{t \rightarrow \infty} s_1(t) = 0$, one can obtain that $\lim_{t \rightarrow \infty} \rho(t) = R$ and $\lim_{t \rightarrow \infty} \eta(t) = 0$ are satisfied in turn. In addition, it is easy to obtain $\lim_{t \rightarrow \infty} g(t) = 0$ by combining with $\lim_{t \rightarrow \infty} s_2(t) = 0$. Thus, there holds $\lim_{t \rightarrow \infty} \hat{\alpha}(t) = d$ from the second equation in (25). Finally, one can get $\lim_{t \rightarrow \infty} f(t) = 0$ and $\lim_{t \rightarrow \infty} \omega(t) = \Omega \mathbf{1}_n$ in a similar trace to the proof of Theorem 1. The above developments imply that the agents asymptotically reach the desired formations. This completes the proof. $\square$

Remark 4: Different from the proposed protocol (12) associated with the condition $l_i > n\bar{a}$, $i = 1, 2$, the adaptive protocol (21) can be carried out without the knowledge of the upper bound $\bar{a}$ of target's acceleration in advance any more.

IV. SIMULATION EXAMPLES

In this section, several simulations are performed to show the effectiveness of the designed formation controllers (12) and (21). An MAS consisting of six agents is considered and their initial states are generated randomly. For highlighting the flexible applications for different requirements, three scenarios are provided in the following.

Example 1: The target moves on a circle around the coordinate origin with the uniform angular velocity of 0.12 and the radius of 20, and its initial position is $[0, 20]^T$ in a plane. It is easy

TABLE I
THREE FORMATIONS IN EXAMPLES 1 AND 3

Formations	d	R	Ω
Linear formation	$[0.1244, 0.4636, 1.5708, 0.4636, 0.1244, 3.5364]^T$	$[5.0990, 3.1622, 1.4142, 1.4140, 3.1622, 5.0990]^T$	-1
Triangular formation	$[0.1799, 0.9273, 0.9273, 0.1799, 2.0344, 2.0344]^T$	$[2.0000, 4.4722, 2.6834, 4.4722, 2.0000, 0.8944]^T$	0
Circular formation	$[\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}]^T$	$[4, 4, 4, 4, 4, 4]^T$	2

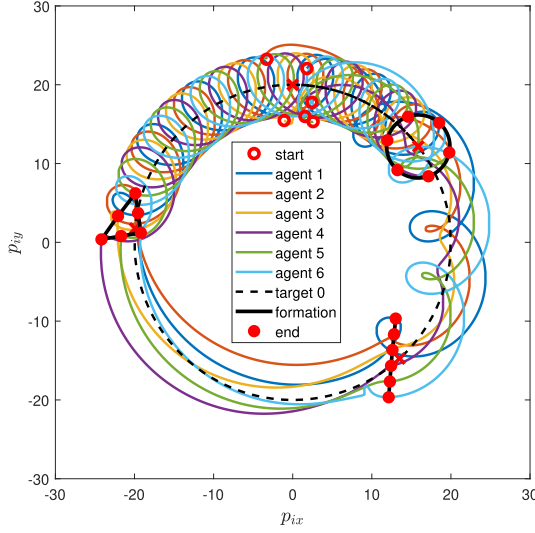


Fig. 3. Position evolution $p_i(t) = [p_{ix}(t), p_{iy}(t)]^T$ of each agent and the target in Example 1.

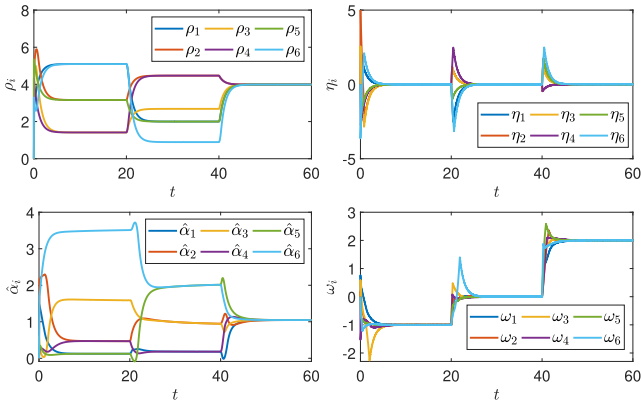


Fig. 4. State evolution of each agent in Example 1.

to acquire that the acceleration $a_0(t)$ satisfies $\|a_0(t)\|_2 \leq 0.288$, that is, $\bar{a} = 0.288$, then the control parameters in the protocol (12) can be selected as $l_1 = l_2 = 7$. The switching desired formations are set in Table I. From Fig. 3, it can be seen that six agents form and maintain various desired formations. The corresponding state trajectories are illustrated in Fig. 4.

Example 2: The initial states of the target are $p_0(0) = [0, 10]^T$ and $v_0(0) = [10, 0]^T$, respectively. The motion trajectories of the target are divided into three stages: 1) it

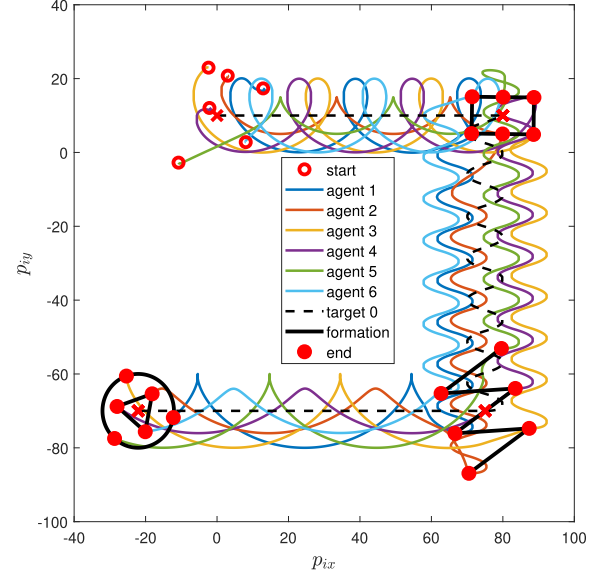


Fig. 5. Position evolution $p_i(t) = [p_{ix}(t), p_{iy}(t)]^T$ of each agent and the target in Example 2.

moves along a straight line with time-varying acceleration $a_0(t) = [2, 0]^T \rightarrow [-2, 0]^T$, where $\rightarrow$ indicates that a_0 switches from the left to the right at some certain time instant; 2) it traces a sine wave trajectory with $a_0(t) = [-5\text{sign}(p_0(t) - 75), -2]^T \rightarrow [-5\text{sign}(p_0(t) - 75), 2]^T$; 3) it tracks along a line with $a_0(t) = -v_0(t) \rightarrow [0, 0]^T \rightarrow [-1.4, 0]^T$. By some calculations, the upper bound of $a_0(t)$ is $\bar{a} = 7.04$, and then, the control parameters in the protocol (12) can be selected as $l_1 = l_2 = 120$. The switching desired formations in the above stages are presented in Table II. Fig. 5 shows that the six agents can achieve and maintain different desired formations with the states of all agents indicated in Fig. 6.

Example 3: The initial position and the trajectory of the target are the same as those in Example 1. Then, each agent i employs the adaptive protocol (21) with the initial values of the adaptive parameters chosen as $e_{i1} = 0$ and $e_{i2} = 0$ at the beginning of each stage. In addition, chose $\tau_{i1} = 2$ and $\tau_{i2} = 2$ for each agent i . In this case, the switching desired formations are given in Table I. It is observed from Fig. 7 that the agents complete and maintain three desired formations. In addition, Figs. 8 and 9 demonstrate that the state and the adaptive parameters of each agent converge to the corresponding values finally.

TABLE II
THREE FORMATIONS IN EXAMPLE 2

Formations	d	R	Ω
Rectangular formation	$[\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}]^T$	$[10, 5, 10, 10, 5, 10]^T$	0.4
Broken-linear formation	$[0.6830, 1.467, 0.9917, 0.6830, 1.4669, 0.9917]^T$	$[10.4051, 17.5112, 13.2057, 10.4051, 17.5112, 13.2057]^T$	0
Hybrid formation	$[\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}]^T$	$[10, 6, 10, 6, 10, 6]^T$	-0.25

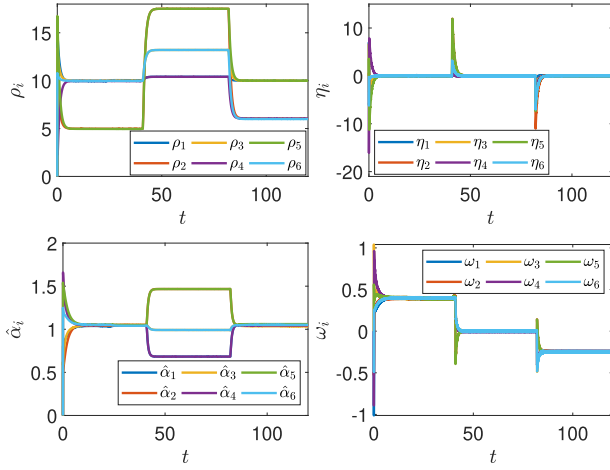


Fig. 6. State evolution of each agent in Example 2.

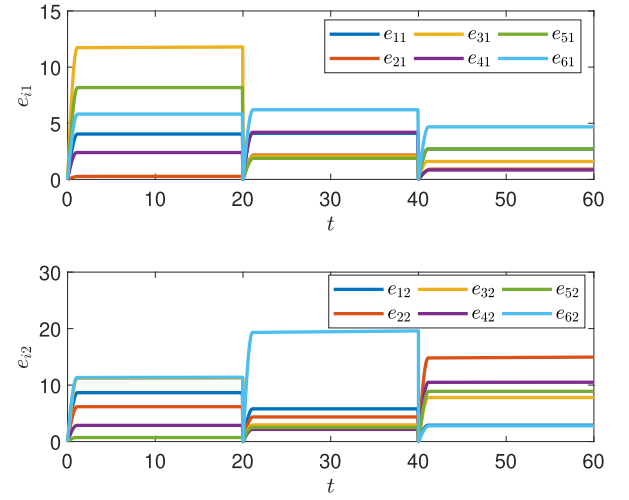


Fig. 8. Adaptive parameter trajectories of the agents in Example 3.

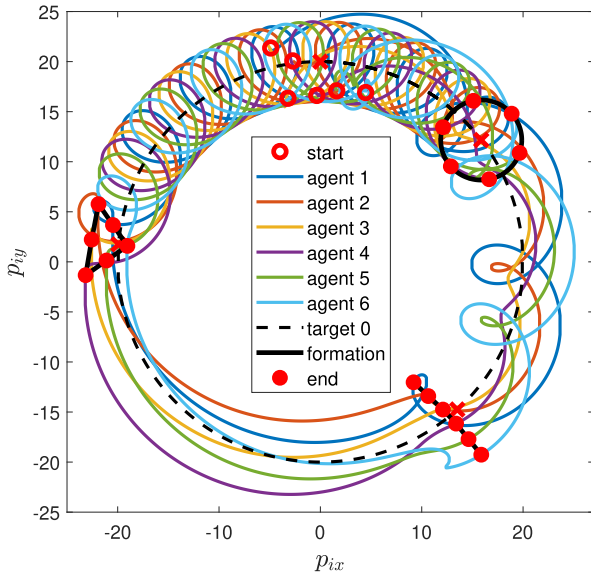


Fig. 7. Position evolution $p_i(t) = [p_{ix}(t), p_{iy}(t)]^T$ of each agent and the target in Example 3.

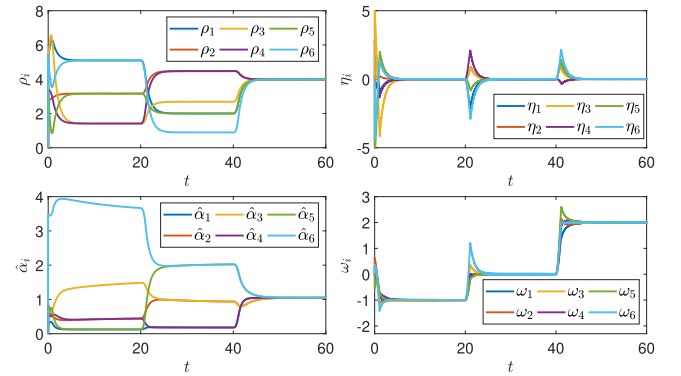


Fig. 9. State evolution of each agent in Example 3.

V. CONCLUSION

This article studied the distributed circle surrounding control problems for double-integrator MASs. Distributed formation control protocols have been proposed to form and maintain the expected arbitrary circle formations in a plane. The proposed protocols do not rely on the target's acceleration information in real time, and thus, can efficiently reduce the requirement of the

equipped devices and data processing capabilities. In addition, the dynamics of each agent described in the local frames were developed by using the transformation relation between the global and local frames. As a result, all the information used in the controller design of each agent can be obtained by measuring or calculating from its local coordinate frame. In conclusion, the proposed design framework can be flexibly employed according to specific requirements. In future, we will focus on extending the results to the case that each agent only can obtain relative position information with the target and its neighbors.

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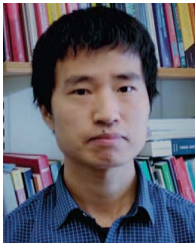
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